Statistical Learning: Classification II – Logistic Regression

Course of Statistical Learning

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Presentation Outline

Limits of Linear Regression for qualitative response

Example: predict medical condition of a patient on the basis of her symptoms

- *Medical condition*: stroke, drug overdose, epilectic seizure
- We could encode the medical condition using a *quantitative* response Y:

$$Y = \begin{cases} 1 & \text{if stroke;} \\ 2 & \text{if drug overdose;} \\ 3 & \text{if epilectic seizure;} \end{cases}$$

- Then use least square to fit a linear regression model to predict Y on the basis of a set of predictors $\{X_1, \ldots, X_p\}$

Quantitative vs. qualitative response

- Observation: the order of the encoding counts!
- We are assuming that drug overdose is between stroke and epilectic seizure
- Moreover, we are assuming that the *distance* between stroke and drug overdose is the same as the difference between drug overdose and epilectic seizure.
- Another reasonable **encoding** would result in totally **different relationships** among the three conditions, and would produce fundamentally **different linear models**:

$$Y = \begin{cases} 1 & \text{if stroke;} \\ 2 & \text{if drug overdose;} \\ 3 & \text{if epilectic seizure;} \end{cases} \qquad Y = \begin{cases} 1 & \text{if epilectic seizure;} \\ 2 & \text{if stroke;} \\ 3 & \text{if drug overdose;} \end{cases}$$

Converting qualitative response to quantitative response

- If the values of the response variable have a natual ordering and the gap between these values are equal, we can use a quantitative encoding

$$Y = \begin{cases} 1 & \text{if mild;} \\ 2 & \text{if moderate;} \\ 3 & \text{if severe;} \end{cases}$$

- However, **in general there is no natural way** to encode a qualitative response variables with more than two values to a quantitative response.

Converting binary qualitative response to quantitative response

- Assume the **medical condition** response variable can take only two values: stroke and drug overdose
- We can use a **binary encoding** (similar to dummy variables for linear regression)

$$Y = \begin{cases} 0 & \text{if stroke} \\ 1 & \text{if drug overdose} \end{cases}$$

- **Fit linear regression** to this binary response
- **Predict** drug overdose if $Y > 0.5$ and stroke otherwise.
- The result of linear regression ($X\hat{\beta}$) is an estimate of $Pr(\text{drug overdose}|X)$.
- However:
  1. the estimate could be **outside** of the $[0, 1]$ interval
  2. **cannot be extended** to qualitative response with **more than two values**
- **We need classification methods suited for classification**

Presentation Outline

Logistic Regression

- Example: Default data set - the response default falls into one of two categories, Yes or No
- Rather than modelling this response Y directly, logistic regression models **the probability that Y belongs to a particular category**
- For example, the probability of default given balance can be written as

$$p(balance) = Pr(default = Yes \,|\, balance)$$

- Then for any given value of balance, a prediction can be made for default. For example, one might predict $default = Yes$ for any individual for whom $p(balance) > 0.5$

Issues with Linear Regression

- We could use a **linear regression model** to represent the relationship between the conditional probability associated to the predictor, $p(X) = Pr(Y = 1 \mid X)$, and the predictor, X

$$p(X) = \beta_0 + \beta_1 X$$

- If we use this approach to predict $p(\mathit{default} = \mathit{Yes})$ using balance, we may run into some **issues**:
  - $P < 0$: for balances close to zero, predicted probability is negative;
  - $P > 1$: for very large balances, we would get P bigger than 1.
- These predictions are **not feasible**, since of course the true **probability** of default, regardless of credit card balance, must fall **between** 0 **and** 1.

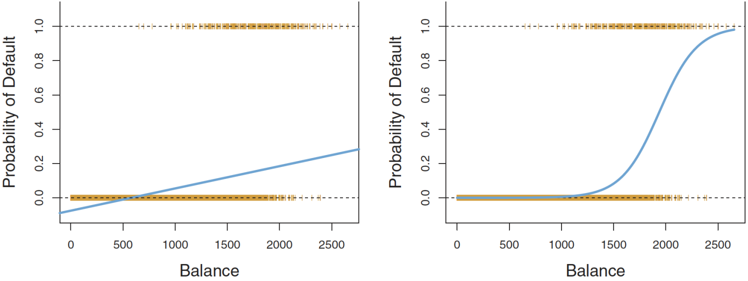
The logistic function

- Any time a **straight line** is fit to a **binary response** that is coded as 0 or 1, in principle **we can predict** $p(X) < 0$ for some values of X **and** $p(X) > 1$ for others (unless the range of X is limited)
- To **avoid this problem**, we must model $p(X)$ using a function that gives outputs between 0 and 1 for all values of X
- In **logistic regression**, we use the **logistic function**

$$p(X) = \frac{e^{\beta_0+\beta_1 X}}{1 + e^{\beta_0+\beta_1 X}}$$

- To fit this model, we use a method called **maximum likelihood**
- The logistic function will always produce an **S-shaped curve** of this form, and so regardless of the value of X, we will obtain a **reasonable prediction**.

Probability of default vs balance



Left: the **estimated probability** of default are computed using **linear regression**. It turns out that some estimated probabilities are **negative**! **Right**: predicted probabilities of default using **logistic regression**, so that all probabilities lie between 0 and 1.

The odds

- After a bit of manipulation of the equation, we find that

$$\frac{p(X)}{1-p(X)} = e^{\beta_0+\beta_1 X}$$

- The quantity $\frac{p(X)}{1-p(X)}$ is called the **odds**, and ranges in $[0, \infty)$
- For example, **on average** 1 **in** 5 **people with an odds of** $\frac{1}{4}$ **will default**, since $p(X) = 0.2$ implies an odds of $\frac{0.2}{1-0.2} = \frac{1}{4}$.
- Likewise on average **nine out of every ten people with an odds of** 9 **will default**, since $p(X) = 0.9$ implies an odds of 9.

Logit or log-odds

- By taking the **logarithm** of both sides, we have

$$log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X$$

where the left-hand side is called **logit** or **log-odds**. Hence, the logistic regression model has a logit that is **linear in** X

- In a logistic regression model, increasing X by **one unit** changes the **log odds** by β_1, or equivalently it multiplies the odds by e^{β_1}

- The amount that $p(X)$ changes due to a **one-unit change** in X will depend on the current value of X. But regardless of the value of X, if β_1 is **positive** then increasing X will be associated with increasing $p(X)$, and if β_1 is **negative** then increasing X will be associated with decreasing $p(X)$

Estimating the Regression Coefficients

- The coefficients β_1 and β_0 must be **estimated** from **training data**
- We could use (non-linear) least squares but the more general **maximum likelihood** is preferred, since it has better statistical properties
- **Intuition:** we seek **estimates** for β_0 and β_1 such that the **predicted probability** $\hat{p}(x_i)$ of **default** for each individual, corresponds **as closely as possible** to the individual's **observed** default status
- In other words, we try to find $\hat{\beta}_0$ and $\hat{\beta}_1$ such that plugging these estimates into the model for $p(X)$ yields a number **close to one** for all individuals who **defaulted**, and a number **close to zero** for all individuals who did **not**

The Likelihood Function

This intuition can be formalized using a mathematical equation called a **likelihood function**:

$$\ell(\beta_0, \beta_1) = \prod_{i:y_i=1} p(x_i) \prod_{i':y_{i'}=0} (1 - p(x_{i'}))$$

- The estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ are chosen to **maximize this likelihood function**
- In the linear regression setting, the **least squares** approach is in fact a **special case of maximum likelihood**. In general, logistic regression and other models can be easily fit using a **statistical software package**

Estimating the Coefficents for the Default dataset

- The table shows the **coefficient estimates** and related information that result from fitting a **logistic regression model** on the Default data in order to predict the probability of *default* = *Yes*, using balance.

	Coefficient	Std. error	Z-statistic	P-value
`Intercept`	-10.6513	0.3612	-29.5	<0.0001
`balance`	0.0055	0.0002	24.9	<0.0001

- $\hat{\beta}_1 = 0.0055$ hence a **one-unit increase** in balance implies an increase in the **log odds** of default by 0.0055 units.
- Many aspects of **logistic regression** are **similar** to **linear regression**. For example, the **accuracy** of coefficient estimates can be measured by **standard errors**. The **z-statistic** (analogous of the t-statistic in linear regression), associated with β_1, is equal to $\hat{\beta}_1/SE(\hat{\beta}_1)$, hence a **large** (absolute) value of the z-statistic indicates **evidence against the null hypothesis** $H_0 : \beta_1 = 0$

Making Predictions

- Once the **coefficients** are estimated, it is simple to compute the probability of default for any credit card balance.

- **Example:** using the coefficient estimates in the table, the default probability for an individual with a **balance of** \$1000 is predicted as

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0+\hat{\beta}_1 X}}{1+e^{\hat{\beta}_0+\hat{\beta}_1 X}} = \frac{e^{-10.6513+0.0055\times 1{,}000}}{1+e^{-10.6513+0.0055\times 1{,}000}} = 0.00576\,,$$

  which is below 1%.

- In contrast, the predicted **probability of default** for an individual with a **balance of** \$2000 is much **higher**, and equals $0.586 = 58.6\%$.

Using qualitative predictors

It is possible to use **qualitative predictors** in a logistic regression model

- The Default dataset includes the qualitative variable Student
- Encode student with a dummy variable: 0: non-student, 1: student

	Coefficient	Std. error	z-statistic	p-value
`Intercept`	-3.5041	0.0707	-49.55	<0.0001
`student[Yes]`	0.4049	0.1150	3.52	0.0004

Positive coefficient associated with the dummy variable (Student[yes]) indicates that students have **higher chance** of default

Multiple Logistic Regression

- **Problem:** predicting a binary response using **multiple predictors**. By analogy with the extension from **simple** to **multiple** linear regression, we can generalize the logistic equation as follows:

$$\log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p$$

where $X = (X_1, \ldots, X_p)$ are p predictors

- The equation can be rewritten as

$$p(X) = \frac{e^{\beta_0+\beta_1 X_1+\ldots+\beta_p X_p}}{1+e^{\beta_0+\beta_1 X_1+\ldots+\beta_p X_p}}$$

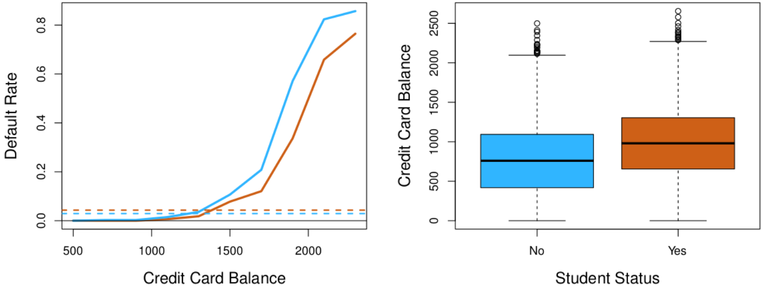
and, then, use the **maximum likelihood** method to estimate $\beta_0, \beta_1, \ldots, \beta_p$

Example of Multiple Logistic Regression

	Coefficient	Std. error	z-statistic	p-value
Intercept	-10.8690	0.4923	-22.08	<0.0001
balance	0.0057	0.0002	24.74	<0.0001
income	0.0030	0.0082	0.37	0.7115
student[Yes]	-0.6468	0.2362	-2.74	0.0062

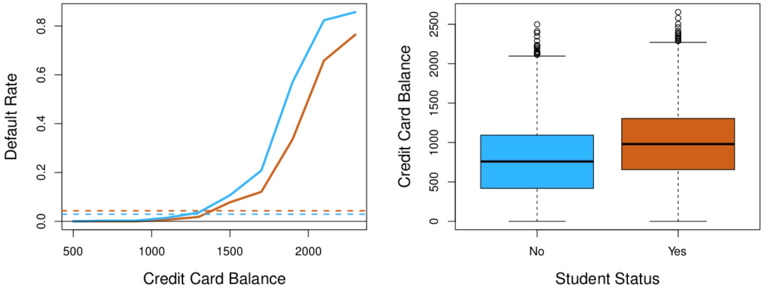
Estimate for a logistic regression model using **3 predictors**: balance, income (thousands of dollars) and student (dummy variable, 0: non-student, 1: student). Negative coefficient associated with the dummy variable (Student[yes]) indicates that students have **lower** chance of default

Confounding effect for the Default dataset



Left: Solid orange (blue) line indicates default rate for students (non-students) as a function of balance. **Dashed orange (blue) line** indicates default rate for students (non-students) averaged over balance and income.
For a **fixed** value of balance students are less likely to default than non-students. However, the **overall** default rate for students is higher than non-students

Confounding effect for the Deafult dataset



Right: The vairables student and balance are **correlated**: **students tend to have higher balance**. Since balance is (positively) correlated with probability of default students **overall** have higher chance of default. However, **given a credit card balance** a student is less likely to default than a non-student. This is a very **important information for a credit card company**, but it would be lost if we perform logistic regression with a single predictor.

Logistic Regression with more than two classes

- We sometimes wish to classify a response variable that has **more than two classes**.
- The two-class logistic regression models discussed in the previous sections have **multiple-class extensions**, but **in practice they tend not to be used all that often**.
- One of the reasons is that the method we discuss in the next section, **Linear Discriminant Analysis**, is **popular for multiple-class classification**.